OMAR NAGY

DATA ENGINEER

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PROFILE

Computer Engineering and Automatic Control graduate building Python and SQL data workflows, ETL pipelines, applied machine learning services, and retrieval-backed applications. Recent work spans a bilingual text-to-SQL system, model serving, data validation, and cloud deployment.

CORE SKILLS

Data engineering: Python, SQL, ETL, data modeling, DataOps, data validation, schema discovery, SQLite, PostgreSQL, MySQL, SQL Server, Oracle
Analytics and ML: Pandas, NumPy, scikit-learn, XGBoost, EDA, feature engineering, model evaluation
Platforms and development: LangGraph, FastAPI, Flask, React, Next.js, Docker, GitHub Actions, Azure Container Apps, Git, REST APIs, RAG

EXPERIENCE AND TRAINING

DataOps Intern Oct 2026 - Jan 2027 | Ongoing
National Telecommunication Institute (NTI) HireReady | Internship & practical training
- Four-month, industry-aligned program with 120 hours of in-company practical training.
- Built Python and SQL workflows and ETL pipelines to prepare data from multiple sources.
- Added automated data cleaning and validation checks to improve consistency.
- Designed database structures and data models, with scheduling, monitoring, and error handling for workflows.

Machine Learning Trainee Nov 2025 - Jul 2026
Digital Egypt Pioneers Initiative (DEPI) - Microsoft Track | Training program
- Applied data preprocessing, feature engineering, and model evaluation to machine learning tasks.
- Built experience with supervised and unsupervised learning workflows.

Machine Learning Trainee Aug 2025 - Sep 2025
Creativa Innovation Hub | Training program
- Performed exploratory data analysis and feature engineering.
- Used cross-validation and hyperparameter tuning to compare model performance.

SELECTED PROJECTS

DataPilot AI | Agentic Text-to-SQL Data Analyst
A bilingual English and Arabic data analyst that turns natural-language questions into safe SQL, query results, visualizations, and insights.

HousePrice AI | Smart Real Estate Valuation
An end-to-end housing price prediction application with an XGBoost model, Flask REST API, and Next.js frontend.

Medical Question Answering Chatbot | Retrieval-Augmented Generation
A medical Q&A prototype that retrieves relevant passages from medical PDFs before generating a response.

EDUCATION

Bachelor of Science in Computer Engineering and Automatic Control | Faculty of Engineering, Tanta University, Tanta, Egypt | 2026 | Excellent with Honors

CERTIFICATION

Programming Fundamentals Diploma | Route IT Training Center | 17 December 2023